

Algorithmic Feed Simulation Classroom Run Kit

TOK Cognitive Science Activity Bank • Knowledge and Technology • 30-40 min

Category	Knowledge and Technology	Time	30-40 min
Difficulty	Medium	ID	algorithmic-feed-simulation

Big idea: A feed is a knowledge filter.

One-Minute Teacher Brief

Students experience how personalisation filters knowledge.

Student Prompt

Inspect Algorithmic Feed Simulation Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Algorithmic Feed Simulation Stimulus A	Student-facing stimulus 1: Paper post cards ranked by engagement, recency, outrage, or similarity to past clicks. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Algorithmic Feed Simulation Stimulus B	Student-facing stimulus 2: A feed that changes after each student 'likes' or skips a card. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Use this for comparison, a second round, or a partner challenge.
Algorithmic Feed Simulation Reveal / Twist	Student-facing stimulus 3: A comparison between a curiosity-maximising feed and a reliability-maximising feed. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Teacher reveal: connect the changed response to the big idea — A feed is a knowledge filter.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Algorithmic Feed Simulation Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Algorithmic Feed Simulation Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Do algorithms expand knowledge or narrow it?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Knowledge and Technology.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how a feed is a knowledge filter.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: Do algorithms expand knowledge or narrow it?

Setup Checklist

- Prepare a starter stimulus using: Paper post cards ranked by engagement, recency, outrage, or similarity to past clicks.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Example Stimuli or Materials

- Paper post cards ranked by engagement, recency, outrage, or similarity to past clicks.
- A feed that changes after each student 'likes' or skips a card.
- A comparison between a curiosity-maximising feed and a reliability-maximising feed.

Resources To Use

- Resource pack: <resources/algorithmic-feed-simulation/index.html>
- Card deck CSV: <resources/algorithmic-feed-simulation/cards.csv>
- Visual prompt: <resources/algorithmic-feed-simulation/visual-prompt.svg>
- Printable student worksheet with observation, inference, evidence, and confidence columns.
- Teacher notes with setup checklist, sample facilitation language, misconceptions, and assessment look-fors.
- Classroom run kit with prompt cards, example stimuli, and debrief moves.
- PowerPoint deck with a student-facing sequence for running the activity.

Run Sequence

1. Students secretly choose five cards they would click.
2. The teacher acts as the algorithm and gives each student more cards similar to their choices.
3. Repeat for three rounds.
4. Students compare how different feeds became over time.

Debrief Moves

- Knowledge question: Do algorithms expand knowledge or narrow it?
- Is a personalised feed a window, a mirror or a cage?
- Who is responsible for algorithmic bias: user, designer, company or society?

Teacher Quick Script

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...